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Surgical planning guidance and learning

Abstract

A method for improving the development of an initial surgical plan includes receiving input information, developing an initial surgical plan based upon the input information, and allowing a user to customize the initial surgical plan by providing input related to modifications to the initial surgical plan. The method further includes storing information related to the modifications to the initial surgical plan and using the stored information to develop a subsequent initial surgical plan based on the modifications.

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Background/Summary

CROSS-REFERENCE TO RELATED PATENT APPLICATION (1) This application is a continuation of U.S. patent application Ser. No. 15/894,461, filed Feb. 12, 2018, which is a continuation of U.S. patent application Ser. No. 14/145,619, filed Dec. 31, 2013, which claims the benefit of and priority to U.S. Provisional Patent Application No. 61/747,765, filed Dec. 31, 2012, each of which is incorporated by reference herein in its entirety.

BACKGROUND

(1) The present invention relates generally to surgical planning in connection with computer-assisted surgeries. More particularly, the embodiments described herein relate to the interactions between a user and an interface of a surgical planning system.

(2) Planning systems for computer-assisted surgical systems produce surgical plans based on input information. To develop the surgical plan, the planning system applies an algorithm to the input information. For example, during planning of a total knee arthroplasty, input information may include information related to the patient's bone structure and other physical characteristics. The surgical plan developed by the planning system shows where any implants, such as a femoral component and a tibial component, should be placed.

(3) After the surgical plan has been produced, the user (e.g., a surgeon, other medical practitioner, or a technical specialist) can customize the plan based on the user's additional knowledge. However, in conventional planning systems, the user may not know how the planning system arrived at the surgical plan. In other words, the user does not understand the algorithm applied to the input information to develop the surgical plan. A lack of knowledge related to the underlying algorithm used by the planning system may make it more difficult for the user to effectively modify the surgical plan (e.g., the planned implant placement or other specifics related to the surgical plan).

SUMMARY

(4) The methods described herein guide a user during surgical planning in a manner that provides the user with information related to an initial surgical plan, thereby improving the transparency of the planning system relative to conventional planning systems. In addition, the methods for guiding a user described herein allow the user to customize the initially developed surgical plan.

(5) In accordance with one aspect, the present disclosure relates to a method for guiding a user during surgical planning. The method includes receiving input information; developing an initial surgical plan based upon the input information; and guiding a user by providing suggested actions to the user. If the user performs the suggested actions, the suggested actions lead the user to the initial surgical plan. The method further includes providing the user with an option to deviate from one or more of the suggested actions by performing non-suggested actions, wherein deviation from one or more of the suggested actions leads to development of a final surgical plan that is different from the initial surgical plan.

(6) According to another aspect, the present disclosure relates to a system for guiding a user during surgical planning. The system includes a processing circuit configured to receive input information; apply an algorithm to the input information to develop an initial surgical plan; and display on a display device a plurality of icons in sequence, each of the plurality of icons having a distinguishing characteristic configured to alert the user to select each of the plurality of icons. If the user selects each of the plurality of icons having a distinguishing characteristic, the selections lead the user to the initial surgical plan. The processing circuit is further configured to customize the initial surgical plan to create a final surgical plan upon user selection of a displayed icon without a distinguishing characteristic.

(7) The invention is capable of other embodiments and of being practiced or being carried out in

various ways. Alternative exemplary embodiments relate to other features and combinations of features as may be generally recited in the claims.

Description

BRIEF DESCRIPTION OF THE FIGURES

(1) The invention will become more fully understood from the following detailed description, taken in conjunction with the accompanying drawings, wherein like reference numerals refer to like elements, in which:

(2) FIG. **1**A is a representation of a display screen during a first step of guiding a user during surgical planning according to a first exemplary embodiment.

(3) FIG. **1**B is a representation of a display screen during a second step of guiding a user during surgical planning according to the first exemplary embodiment.

(4) FIG. **1**C is a representation of a display screen during a third step of guiding a user during surgical planning according to the first exemplary embodiment.

(5) FIG. **1**D is a representation of a display screen during a fourth step of guiding a user during surgical planning according to the first exemplary embodiment.

(6) FIG. **1**E is a representation of a display screen during a fifth step of guiding a user during surgical planning according to the first exemplary embodiment.

(7) FIG. **1**F is a representation of a display screen after completion of guiding a user during surgical planning according to the first exemplary embodiment.

(8) FIG. **2**A is a representation of a display screen during a first step of guiding a user during surgical planning according to a second exemplary embodiment.

(9) FIG. **2**B is a representation of a display screen during a second step of guiding a user during surgical planning according to the second exemplary embodiment.

(10) FIG. **2**C is a representation of a display screen after completion of guiding a user during surgical planning according to the second exemplary embodiment.

(11) FIG. **3**A is a representation of a display screen during a first step of guiding a user during surgical planning according to a third exemplary embodiment.

(12) FIG. **3**B is a representation of a display screen during a second step of guiding a user during surgical planning according to the third exemplary embodiment.

(13) FIG. **3**C is a representation of a display screen during a third step of guiding a user during surgical planning according to the third exemplary embodiment.

(14) FIG. **3**D is a representation of a display screen during a fourth step of guiding a user during surgical planning according to the third exemplary embodiment.

(15) FIG. **3**E is a representation of a display screen during a fifth step of guiding a user during surgical planning according to the third exemplary embodiment.

(16) FIG. **3**F is a representation of a display screen during a sixth step of guiding a user during surgical planning according to the third exemplary embodiment.

(17) FIG. **3**G is a representation of a display screen after completion of guiding a user during surgical planning according to the third exemplary embodiment.

(18) FIG. **4**A is a representation of a display screen during a first step of guiding a user during surgical planning according to a fourth exemplary embodiment.

(19) FIG. **4**B is a representation of a display screen during a second step of guiding a user during surgical planning according to the fourth exemplary embodiment.

(20) FIG. **4**C is a representation of a display screen during a third step of guiding a user during surgical planning according to the fourth exemplary embodiment.

(21) FIG. **4**D is a representation of a display screen during a fourth step of guiding a user during surgical planning according to the fourth exemplary embodiment.

(22) FIG. **4**E is a representation of a display screen during a fifth step of guiding a user during surgical planning according to the fourth exemplary embodiment.
(23) FIG. **4**F is a representation of a display screen after completion of guiding a user during surgical planning according to the fourth exemplary embodiment.
(24) FIG. **5** is a perspective view of an embodiment of a surgical system according to an exemplary embodiment.
(25) FIG. **6** is a block diagram of a model surgical system according to an exemplary embodiment.

DETAILED DESCRIPTION

(26) Before turning to the figures, which illustrate the exemplary embodiments in detail, it should be understood that the application is not limited to the details or methodology set forth in the description or illustrated in the figures. It should also be understood that the terminology is for the purpose of description only and should not be regarded as limiting.
(27) FIGS. **1**A-**4**F provide various representations of a user display during different exemplary embodiments described herein. In each of the exemplary embodiments, input information has been received (e.g., by a surgical planning system) and an algorithm has been applied to the input information to develop an initial surgical plan. Input information may relate to the patient's bone structure or other physical characteristics. Imaging techniques, such as CT, MRI, or ultrasound, may be used to obtain three-dimensional representations of the patient's anatomy, such as the representations of the femur and tibia shown in FIGS. **1**A-**4**F. General surgical planning and navigation, including haptic control and feedback, may be performed by a computerized surgical system such as that depicted in FIGS. **5** and **6** (and described below), and as described in U.S. Pat. No. 8,010,180 “Haptic Guidance System and Method” to Quaid et al., which is incorporated herein by reference in its entirety.
(28) FIGS. **1**A-**4**F provide examples of guiding a user during surgical planning by providing a plurality of suggested actions to the user. The user is guided via a display screen, shown in various exemplary embodiments as graphical user interface (GUI) **2**, to perform the suggested actions. The suggested actions may correspond to steps of an algorithm used by a planning system to develop an initial surgical plan. Guiding a user in this manner allows the user to 1) learn how the initial surgical plan was developed by the planning system, and 2) customize the initial surgical plan during the user-guidance process. If the user performs the suggested actions, the suggested actions will lead the user to the initial surgical plan. However, the user has the option to deviate from one or more of the suggested actions by performing non-suggested actions. Deviation from one or more of the suggested actions leads to development of a final surgical plan that is different from the initial surgical plan.
(29) In connection with guiding a user, the GUI **2** illustrated in FIGS. **1**A-**4**F may display a series of icons in sequence, each of the icons having a distinguishing characteristic intended to alert the user to select the icons. If the user selects the icons having a distinguishing characteristic, the selections will lead the user to the initial surgical plan. However, the user can customize the initial surgical plan by selecting a displayed icon without a distinguishing characteristic.
(30) In accordance with one embodiment, GUI **2** provides virtual representations of a femoral component **4** and a tibial component **6** on virtual models of a patient's femur and tibia, respectively. References herein to femoral component **4** and tibial component **6** are understood to be references to the virtual representations of these components as shown on a GUI **2**.
(31) The GUI **2** also displays criteria in the form of a list **8**. The criteria are related to the surgical plan shown on the current display of GUI **2**. Each of the criteria is associated with one or more acceptable values. Acceptable values for any particular criterion may be values below a maximum threshold, above a minimum threshold, or within a predefined range. Some examples of displayed criteria, as can be seen in FIGS. **1**A-**4**F, include femoral notching and sizing, femoral rotation, femoral overhang, tibial overhang and sizing, tibial rotation, tibial slope, limb alignment and joint line, extension and flexion gaps, and mid-flexion gap and stability. These criteria relate to surgical

planning of a total knee arthroplasty (the example used in the exemplary embodiments of FIGS. **1**A-**4**F). However, the criteria may be chosen to be applicable to any planned surgical procedure.

(32) Each of the criteria in list **8** includes a corresponding indicator **10** configured to provide the user with information related to the particular criterion. The information provided by indicator **10** may relate to whether the corresponding criterion has an acceptable value. The indicators **10** may be circular images configured to provide information via a display feature. Each indicator **10** may be configured to change its display feature to convey information. In one embodiment, if a certain criterion has an acceptable value during a stage of user-guidance, the corresponding indicator indicates acceptability by displaying a first display feature, such as a checkmark. For example, in FIG. **1**A, femoral overhang and several other criteria have acceptable values and are therefore marked with an indicator **10***a* having a first display feature (e.g., a checkmark). However, if a certain criterion does not have an acceptable value during a stage of user-guidance, the corresponding indicator will display a second display feature, such as being filled in. In FIG. **1**A, the femoral notching and sizing criterion and the extension and flexion gaps criterion do not have acceptable values. These criteria are therefore marked with a indicator **10***b* having a second display feature (e.g., a filled interior). Criteria that are within a predefined range of an acceptable value may be marked with an indicator having a third display feature, such as a shaded interior. In FIG. **1**A, the tibial slope criteria is within a predefined range of an acceptable value, so the indicator **10***c* associated with tibial slope has a shaded interior. Thus, the indicators **10***a*, **10***b*, and **10***c* can provide information related to the current values of the criteria in list **8** by displaying different features. Any type of display features may be utilized to convey information to a user.

(33) In one embodiment, the indicators **10** provide information related to changes in the values of the criteria in list **8** by changing color in connection with changes to the criteria values. Thus, the first, second, and third display features can be different colors. The indicators **10** may change color when the user performs suggested or non-suggested actions that change one or more of the criteria. A change of a criterion value may be, for example, a change from an unacceptable value to a value within a certain range of an acceptable value. The corresponding indicator indicates this change by changing from an indicator **10***b* having a second color (e.g., red) to an indicator **10***c* having a third color (e.g., yellow). Alternatively, a change in criterion value may be a change from a value within a certain range of an acceptable value to an acceptable value. The corresponding indicator indicates this change by changing from an indicator **10***c* having a third color (e.g., yellow) to an indicator **10***a* a having a first color (e.g., green).

(34) GUI **2** further includes a plurality of icons **12***a*-**12***j*. A user may select (e.g., touch, mouse click, etc.) the icons to cause additional information to be displayed. Various types of information that may be displayed to a user include or relate to: 3D flex pose capture (icon **12***a*), limb alignment (icon **12***b*), gap graphs (icons **12***c* and **12***h*), virtual models of implants (icon **12***d*), planned bone resections (icon **12***e*), CT scan overlay (icon **12***f*), alignment axes (icon **12***g*), trace points (icon **12***i*) and over/under hang (icon **12***j*).

(35) FIGS. **1**A-**1**F illustrate a first exemplary embodiment for guiding a user during surgical planning. In the embodiment of FIGS. **1**A-**1**F, the user is guided through steps relating to planning the placement of femoral and tibial components for a total knee arthroplasty. Referring to FIG. **1**A, a GUI **2** during a first step of guiding a user is shown. The GUI **2** provides a first suggested action to a user by displaying an icon having a distinguishing characteristic. The distinguishing characteristic is configured to alert the user to select the icon. In FIG. **1**A, icon **12***h* is marked with two distinguishing characteristics—an outline and a guiding arrow **14**. The outline may be shown in a specific color (e.g., pink), and the guiding arrow **14** may be shown in the same or different color. The distinguishing characteristic of icon **12***h* guides the user to select the icon **12***h*. Selecting the icon **12***h* causes additional information related to gaps between femoral component **4** and tibial component **6** to be displayed in the form of bar graphs **16**, as shown in FIG. **1**B. FIG. **1**B illustrates GUI **2** during a second step of guiding a user during surgical planning. In FIG. **1**B, an expansion

box **18** includes a distinguishing characteristic (e.g., an outline in a specific color) to guide the user to select the expansion box **18**. Selecting the expansion box **18** causes placement arrows **19** to appear, as shown in FIG. **1**C.

(36) The GUI **2** next guides the user by suggesting how to adjust the placements of the femoral component **4** and tibial component **6** to achieve acceptable criteria values. The suggested action of FIG. **1**C is selecting an icon to cause a change in placement of a virtual implant (in this step, the femoral component **4**). The marked arrow **20** shown in FIG. **1**C includes a distinguishing characteristic (e.g., an outline in a specific color) to guide the user to select the marked arrow **20**. When the user selects the marked arrow **20**, the placement of femoral component **4** shifts from the placement shown in FIG. **1**C to the placement shown in FIG. **1**D.

(37) As can be seen in FIG. **1**D, the shift of femoral component **4** caused the femoral notching and sizing criterion to have an acceptable value. To provide this information to the user, the femoral notching and sizing criterion in FIG. **1**D is marked with an indicator **10***a* having a checkmark (in contrast, a filled-in indicator **10***b* corresponded to femoral notching and sizing in FIG. **1**C). Colors may also be used in the relevant indicators to provide this information. The user can therefore see a direct cause and effect between shifting the placement of femoral component **4** and the femoral notching and sizing criterion having an acceptable value.

(38) In FIG. **1**D, expansion box **18** includes a distinguishing characteristic (e.g., an outline in a specific color) to guide the user to select expansion box **18**. Selecting expansion box **18** causes placement arrows **19** to appear as shown in FIG. **1**E, allowing the user to adjust the placement of a virtual implant (in this step, tibial component **6**). The user is guided to adjust the posterior slope of the tibial component **6** by selecting marked arrow **24**, which includes a distinguishing characteristic. Selecting marked arrow **24** adjusts the tibial component **6** to the placement shown in FIG. **1**F. The change in placement or orientation may be accomplished incrementally via several selections of arrow **24**. As shown in FIG. **1**F, this adjustment to the posterior slope (e.g., from 2.0° to 7.0°) caused both the tibial slope criterion and extension and flexion gaps criterion to have acceptable values, as shown by their associated checked indicators **10***a*. The GUI **2** indicates that the planning of femoral and tibial component placement is complete by displaying a completion icon **26**.

(39) The actions suggested to the user (e.g., selecting marked arrows **20** or **24**) are presented in a predetermined manner. If the user performs each suggested action of FIGS. **1**A-**1**F (e.g., selects each of the plurality of icons having a distinguishing characteristic), the actions will lead the user to the initial surgical plan developed by applying an algorithm to input information. Thus, via GUI **2**, the user is guided through steps that, if followed, would lead the user to the initially developed surgical plan. This guidance illustrates to the user how the algorithm was applied to arrive at the initial surgical plan. The user guidance therefore provides transparency into how the initial surgical plan was developed.

(40) However, the user has the option to deviate from one or more of the suggested actions by performing non-suggested actions, such as selecting a displayed icon without a distinguishing characteristic. Deviation from one or more of the plurality of suggested actions may lead to the development of a final surgical plan that is different from the initial surgical plan. In this manner, the user can customize the initial surgical plan to create a final surgical plan. The final surgical plan can differ from the initial surgical plan in one or more ways, such as in placement of a virtual implant, type of virtual implant, or size of virtual implant. For example, in the step shown by the GUI **2** in FIG. **1**C, one way the user can customize the initial surgical plan is by selecting the arrows surrounding marked arrow **20**, which do not have distinguishing characteristics. Selecting an arrow other than marked arrow **20** would customize the surgical plan by causing the femoral component **4** to shift to a different placement than the placement shown in FIG. **1**D (which resulted from selecting marked arrow **20**). The user guidance therefore allows the user to customize the initial plan in a straightforward and intuitive manner.

(41) FIGS. **2**A-**2**C illustrate a second exemplary embodiment for guiding a user during surgical planning. FIG. **2**A shows GUI **2** during a first step of guiding the user and illustrates a suggested action of selecting an icon to cause a change in size of a virtual implant. In other embodiments, the suggested action can be selecting an icon to cause a change in type of a virtual implant, such as femoral component **4** or tibial component **6**. Specifically, the user is guided in FIG. **2**A to select box **28**, which has distinguishing characteristics of an outline and a guiding arrow **14**. Selecting box **28** decreases the size of the femoral component **4**. As can be seen in FIG. **2**B, the femur size has been decreased from 6w (FIG. **2**A) to 6N (FIG. **2**B). In FIG. **2**B, the user is guided to select icon **12***j*, which causes information related to over/under hang of femoral component **4** and tibial component **6** to appear in the form of outlines **30** (FIG. **2**C). The user can therefore see how changing the size of femoral component **4** caused femoral overhang (a criterion shown in list **8** that had an unacceptable value in FIG. **2**A) to have an acceptable value (as shown by the indicator **10***a* corresponding to femoral overhang in FIGS. **2**B and **2**C). The outline **30** associated with femoral component **4** in FIG. **2**C shows the user that femoral component **4** is within the outline **30** in the medial and lateral directions. The user can use the outline **30** to evaluate whether he or she agrees with the suggested implant size. FIG. **2**C also includes a completion icon **26** to demonstrate the completion of user guidance.
(42) As in the first exemplary embodiment, although the user's actions (e.g., selecting box **28** to decrease the size of femoral component **4**) in the embodiment of FIGS. **2**A-**2**C are guided in a predetermined manner, the user has the option during each step of the process to adjust the initial surgical plan. For example, in the step shown by the GUI **2** in FIG. **2**A, the user could alternatively decide to shift the femoral component **4** in different directions to eliminate femoral overhang instead of making the femoral component **4** smaller. Alternatively, the user could decide that the amount of femoral overhang shown in the step of FIG. **2**A is acceptable and decide not to adjust the placement or size of femoral component **4**. One of these alternative decisions by the user (e.g., any decision other than following the suggested action of decreasing the size of femoral component **4**) would result in a final plan that is different from the initial surgical plan. In other words, the user is able to customize the initial surgical plan by deciding to take actions different from those suggested by the GUI **2**.
(43) FIGS. **3**A-**3**G illustrate a third exemplary embodiment for guiding a user during surgical planning. Referring to FIG. **3**A, an indicator **10***b* having a second display feature corresponds to the extension and flexion gaps criterion, indicating that this criterion does not have an acceptable value. In other embodiments, the indicator may change color to indicate whether the associated criterion is acceptable or use other ways of indicating the same to the user. The exemplary embodiment of FIGS. **3**A-**3**G leads the user through steps to adjust the femoral component **4** and tibial component **6** in a manner that will bring the extension and flexion gaps criterion to an acceptable value.
(44) Referring again to FIG. **3**A, expansion box **18** includes distinguishing characteristics of both an outline and a guiding arrow **14** to guide the user to select (e.g., touch, click on, etc.) expansion box **18**. When the user selects expansion box **18**, placement arrows **19** appear (as shown in FIG. **3**B) that will allow the user to adjust the placement of femoral component **4**. The marked arrow **34** includes a distinguishing characteristic to guide the user to select the marked arrow **34**. When the user selects the marked arrow **34**, the placement of femoral component **4** shifts upward to the placement shown in FIG. **3**C. Correspondingly, measurements related to gaps between femoral component **4** and tibial component **6** are altered, as shown by the bar graphs **16**, and the extension and flexion gaps criterion now has an acceptable value (see FIG. **3**C). However, the limb alignment and joint line criterion now has an indicator **10***c* having a third display feature, communicating to the user that the limb alignment and joint line criterion no longer has an acceptable value, but is within predetermined ranges of an acceptable value. The user will next be guided to make adjustments that will bring the limb alignment and joint line criterion to an acceptable value.

(45) Accordingly, in FIG. **3**C, the user is guided to select marked arrow **36**, which shifts the femoral component **4** downward. As shown in FIG. **3**D, the bar graphs **16** indicate a corresponding change in the gaps between femoral component **4** and tibial component **6**. The limb alignment and joint line criterion is now at an acceptable value, as shown by the corresponding indicator **10***a*, but the extension and flexion gaps criterion is no longer at an acceptable value, as shown by the corresponding indicator **10***b*. Therefore, the planning system will guide the user to make adjustments that will bring the extension and flexion gaps criterion to an acceptable value. Accordingly, the user is guided in FIG. **3**D to select marked arrow **38**. Selecting marked arrow **38** causes the tibial component **6** to move downward to the position shown in FIG. **3**E. The indicator corresponding to the extension and flexion gaps criterion changes from an indicator **10***b* having a second display feature to an indicator **10***a* having a third display feature to indicate to the user that the previous action (adjusting the tibial component **6**) has brought the extension and flexion gaps criterion closer to an acceptable value.
(46) FIG. **3**E guides the user to select expansion box **18**, which causes additional placement arrows **19** to appear, as shown in FIG. **3**F. In FIG. **3**F, the user is guided to select marked arrow **40**, which rotates the femoral component **4** to the placement shown in FIG. **3**G. This adjustment of femoral component **4** causes the extension and flexion gaps criterion to have an acceptable value, as indicated by the checked indicator **10***a* corresponding to the extension and flexion gaps criterion. In addition, FIG. **3**G includes a completion icon **26**.
(47) FIGS. **4**A-**4**F illustrate a fourth exemplary embodiment for guiding a user during surgical planning. Referring to FIG. **4**A, list **8** indicates that the extension and flexion gaps criterion and the mid flexion gap and stability criterion do not have acceptable values. Therefore, the embodiment of FIGS. **4**A-**4**F will guide the user to adjust the femoral component **4** and tibial component **6** in a manner that will bring the criteria in list **8** to acceptable (or close to acceptable) values.
(48) In FIG. **4**A, icon **12***b* has been selected to display information related to limb alignment in the form of a diagram **44**. The user is guided in FIG. **4**A to select expansion box **18**, which includes the distinguishing features of an outline and a guiding arrow **14**. Selecting expansion box **18** causes placement arrows **19** to appear, as shown in FIG. **4**B. In FIG. **4**B, the user is guided to select marked arrow **42**. Selecting marked arrow **42** rotates femoral component **4** from the placement shown in FIG. **4**B to the placement shown in FIG. **4**C. Correspondingly, the measurements related to gaps between femoral component **4** and tibial component **6** are altered, as shown by the bar graphs **16**, and the femoral varus angle has been altered (e.g., from 0.0° to 1.0°). Diagram **44** also illustrates the change in limb alignment between FIG. **4**B and FIG. **4**C. In FIG. **4**C, the extension and flexion gaps criterion and the mid flexion gap and stability criterion are marked with indicators **10***c* having a third display feature, communicating to the user that these criteria are moving closer to acceptable values. The planning system will then guide the user to make adjustments to bring all criteria to acceptable values.
(49) Accordingly, in FIG. **4**C, the planning system guides the user to select marked arrow **46**. Selecting marked arrow **46** rotates tibial component **6** from the placement shown in FIG. **4**C to the placement shown in FIG. **4**D (e.g., incrementally depending upon the number of times marked arrow **46** is selected). Correspondingly, the measurements related to gaps between femoral component **4** and tibial component **6** are altered, as shown by the bar graphs **16**, and the tibial varus angle has been altered (e.g., from 0.0° to 1.0°). Diagram **44** illustrates the further change in limb alignment. In FIG. **4**D, the extension and flexion gaps criterion and the mid flexion gap and stability criterion continue to be marked with indicators **10***c*, showing that the criteria are close, but are not yet at acceptable values. Furthermore, the limb alignment and joint line criterion is now also marked with an indicator **10***c* having a third display feature. Therefore, in FIG. **4**D, the user is guided to select expansion arrow **18** in order to allow the user to again adjust the placement of femoral component **4**. Selecting expansion box **18** causes additional placement arrows **19** to appear as shown in FIG. **4**E.

(50) In FIG. **4**E, the user is guided to select marked arrow **48**, which rotates the femoral component **4** from the placement shown in FIG. **4**E to the placement shown in FIG. **4**F. In FIG. **4**F, the extension and flexion gaps criterion and the mid flexion gap and stability criterion are at acceptable values, as shown by the corresponding indicators **10***a* having a first display feature (e.g., being checked). However, the limb alignment and joint line criterion continues to have a corresponding indicator **10***c* having the third display feature showing that the criterion is close, but not yet at the desired value. Nonetheless, the planning system has arrived at the initial surgical plan created based on the input information, and FIG. **4**F therefore displays a completion icon **26**.
(51) In some conventional surgical planning systems, a surgical plan is created based on input information. The user is then provided with the surgical plan without having insight into the algorithm used by the planning system to develop the surgical plan. In each of the exemplary embodiments described herein, a planning system creates an initial surgical plan based on certain input information. However, in contrast to other conventional systems, the methods described herein guide the user through a series of steps, via GUI **2**, to lead the user to the initial surgical plan. The algorithm used by the planning system therefore becomes transparent to the user, and the user can easily see how the planning system arrived at the initial surgical plan.
(52) This transparency provides users with the opportunity to customize the initial surgical plan in a straightforward manner. A user might desire to customize the surgical plan for a variety of reasons. For example, the user may have additional knowledge related to the specific patient (e.g., the patient's lifestyle or other factors that may affect the outcome of the surgical procedure), which may cause the user to want to modify the surgical plan in a particular way. As another example, the user may have years of experience performing similar surgical procedures, and may wish to modify the plan in a manner that he or she knows will lead to a more successful outcome. The methods described herein allow a user to more readily determine how to implement customizations by providing the user with awareness of the series of steps taken by the planning system, as well as providing the user with an opportunity to modify the surgical plan at each step.
(53) Another advantage of the methods described herein relative to other planning systems is an improved user “buy in.” In conventional planning systems and methods, a user is simply provided with a solution (e.g., a surgical plan) and is not aware of how the system arrived at the solution. However, when a user is taken step-by-step through the process of arriving at an initial surgical plan, as described herein, the user is able to understand the algorithm relied on by the planning system. For example, when a user is guided according to the embodiment of FIGS. **2**A-**2**C, the user will understand that the initial surgical plan includes a femoral component **4** with a size of “6N” in order to obtain an acceptable value for femoral overhang. Similarly, when a user is guided according to the embodiment of FIGS. **3**A-**3**G, the user will understand that the femoral component **4** and tibial component **6** are placed to ensure acceptable values for the extension and flexion gaps criterion (as well as the other criteria in list **8**). Because the user is provided with more knowledge regarding the algorithm applied by the surgical planning system, the user may be more comfortable relying on and implementing the surgical plan provided by the planning system.
(54) Another advantage of the planning systems and methods described herein is the ability to improve the algorithm applied to input information over time by analysis of user input. User input may include, for example, any user actions, such as input of information or modifications or customizations to a surgical plan, taken in connection with surgical planning as described herein. The planning systems can store data related to this user input, which can later be accessed to improve the planning system's algorithm. For example, in the exemplary embodiment of FIGS. **1**A-**1**F, a certain placement of femoral component **4** is suggested by the planning system to achieve the initial surgical plan illustrated in FIG. **1**F. However, if it is found that users typically modify the suggested femoral placement for patients with a certain type of bone structure, the algorithm might be altered to arrive automatically at the modified femoral placement when creating initial surgical plans in the future. In this manner, data obtained by evaluating user customizations can be applied

to improve the planning system's algorithm.
(55) The methods of guiding a user during surgical planning described herein may be implemented using a treatment planning computing system, such as the computing system associated with a RIO® Robotic Arm Interactive Orthopedic System available from MAKO Surgical Corp., Ft. Lauderdale, Florida FIG. **5** shows an embodiment of an exemplary surgical system **50** in which and for which the techniques described above can be implemented. The surgical system **50** includes a computing system **52**, a haptic device **54**, and a tracking system **56**. In operation, the surgical system **50** enables comprehensive, intraoperative surgical planning. The surgical system **50** may also provide haptic guidance to a user (e.g., a surgeon) and/or limits the user's manipulation of the haptic device **54** as the user performs a surgical procedure.
(56) Embodiments of the subject matter, the methods, and the operations described in this specification can be implemented in digital electronic circuitry, or in computer software embodied on a tangible medium, firmware, or hardware, including the structures disclosed in this specification and their structural equivalents, or in combinations of one or more of them. In the embodiment of FIG. **5**, the computing system **52** may include hardware and software for operation and control of the surgical system **50**. Such hardware and/or software is configured to enable the system **50** to perform the techniques described herein. The computing system **52** includes a surgical controller **62**, a display device **64**, and an input device **66**.
(57) The surgical controller **62** may be any known computing system but is preferably a programmable, processor-based system. For example, the surgical controller **62** may include a microprocessor, a hard drive, random access memory (RAM), read only memory (ROM), input/output (I/O) circuitry, and any other known computer component. The surgical controller **62** is preferably adapted for use with various types of storage devices (persistent and removable), such as, for example, a portable drive, magnetic storage, solid state storage (e.g., a flash memory card), optical storage, and/or network/Internet storage. The surgical controller **62** may comprise one or more computers, including, for example, a personal computer or a workstation operating under a suitable operating system and preferably includes a graphical user interface (GUI).
(58) Referring to FIG. **6**, in an exemplary embodiment, the surgical controller **62** includes a processing circuit **70** having a processor **72** and memory **74**. Processor **72** can be implemented as a general purpose processor executing one or more computer programs to perform actions by operating on input data and generating output. The processes and logic flows can also be performed by, and apparatus can also be implemented as, special purpose logic circuitry, e.g., an FPGA (field programmable gate array) or an ASIC (application specific integrated circuit), a group of processing components, or other suitable electronic processing components. Generally, a processor will receive instructions and data from a read only memory or a random access memory or both. Memory **74** (e.g., memory, memory unit, storage device, etc.) comprises one or more devices (e.g., RAM, ROM, Flash-memory, hard disk storage, etc.) for storing data and/or computer code for completing or facilitating the various processes described in the present application. Memory **74** may be or include volatile memory or non-volatile memory. Memory **74** may include database components, object code components, script components, or any other type of information structure for supporting the various activities described in the present application. According to an exemplary embodiment, memory **74** is communicably connected to processor **72** and includes computer code for executing one or more processes described herein. The memory **74** may contain a variety of modules, each capable of storing data and/or computer code related to specific types of functions. In one embodiment, memory **74** contains several modules related to surgical procedures, such as a planning module **740**, a navigation module **742**, a registration module **744**, and a robotic control module **746**.
(59) Alternatively or in addition, the program instructions can be encoded on an artificially generated propagated signal, e.g., a machine-generated electrical, optical, or electromagnetic signal, that is generated to encode information for transmission to suitable receiver apparatus for

execution by a data processing apparatus. A computer storage medium can be, or be included in, a computer-readable storage device, a computer-readable storage substrate, a random or serial access memory array or device, or a combination of one or more of them. Moreover, while a computer storage medium is not a propagated signal, a computer storage medium can be a source or destination of computer program instructions encoded in an artificially generated propagated signal. The computer storage medium can also be, or be included in, one or more separate components or media (e.g., multiple CDs, disks, or other storage devices). Accordingly, the computer storage medium may be tangible and non-transitory.
(60) A computer program (also known as a program, software, software application, script, or code) can be written in any form of programming language, including compiled or interpreted languages, declarative or procedural languages, and it can be deployed in any form, including as a stand-alone program or as a module, component, subroutine, object, or other unit suitable for use in a computing environment. A computer program may, but need not, correspond to a file in a file system. A program can be stored in a portion of a file that holds other programs or data (e.g., one or more scripts stored in a markup language document), in a single file dedicated to the program in question, or in multiple coordinated files (e.g., files that store one or more modules, sub programs, or portions of code). A computer program can be deployed to be executed on one computer or on multiple computers that are located at one site or distributed across multiple sites and interconnected by a communication network.
(61) Generally, a computer will also include, or be operatively coupled to receive data from or transfer data to, or both, one or more mass storage devices for storing data, e.g., magnetic, magneto optical disks, or optical disks. However, a computer need not have such devices. Moreover, a computer can be embedded in another device, e.g., a mobile telephone, a personal digital assistant (PDA), a mobile audio or video player, a game console, a Global Positioning System (GPS) receiver, or a portable storage device (e.g., a universal serial bus (USB) flash drive), to name just a few. Devices suitable for storing computer program instructions and data include all forms of non-volatile memory, media and memory devices, including by way of example semiconductor memory devices, e.g., EPROM, EEPROM, and flash memory devices; magnetic disks, e.g., internal hard disks or removable disks; magneto optical disks; and CD ROM and DVD-ROM disks. The processor and the memory can be supplemented by, or incorporated in, special purpose logic circuitry.
(62) Embodiments of the subject matter described in this specification can be implemented in a computing system that includes a back end component, e.g., as a data server, or that includes a middleware component, e.g., an application server, or that includes a front end component, e.g., a client computer having a graphical user interface or a Web browser through which a user can interact with an embodiment of the subject matter described in this specification, or any combination of one or more such back end, middleware, or front end components. The components of the system can be interconnected by any form or medium of digital data communication, e.g., a communication network. Examples of communication networks include a local area network (“LAN”) and a wide area network (“WAN”), an inter-network (e.g., the Internet), and peer-to-peer networks (e.g., ad hoc peer-to-peer networks).
(63) Referring to the embodiment of surgical system **50** depicted in FIG. **6**, the surgical controller **62** further includes a communication interface **76**. The communication interface **76** of the computing system **52** is coupled to a computing device (not shown) of the haptic device **54** via an interface and to the tracking system **56** via an interface. The interfaces can include a physical interface and a software interface. The physical interface of the communication interface **76** can be or include wired or wireless interfaces (e.g., jacks, antennas, transmitters, receivers, transceivers, wire terminals, etc.) for conducting data communications with external sources via a direct connection or a network connection (e.g., an Internet connection, a LAN, WAN, or WLAN connection, etc.). The software interface may be resident on the surgical controller **62**, the

computing device (not shown) of the haptic device **54**, and/or the tracking system **56**. In some embodiments, the surgical controller **62** and the computing device (not shown) are the same computing device. The software may also operate on a remote server, housed in the same building as the surgical system **50**, or at an external server site.

(64) Computer system **52** also includes display device **64**. The display device **64** is a visual interface between the computing system **52** and the user. GUI **2** described according to the exemplary embodiments of FIGS. **1**A-**4**F may be displayed on display device **64**. The display device **64** is connected to the surgical controller **62** and may be any device suitable for displaying text, images, graphics, and/or other visual output. For example, the display device **64** may include a standard display screen (e.g., LCD, CRT, OLED, TFT, plasma, etc.), a touch screen, a wearable display (e.g., eyewear such as glasses or goggles), a projection display, a head-mounted display, a holographic display, and/or any other visual output device. The display device **64** may be disposed on or near the surgical controller **62** (e.g., on the cart as shown in FIG. **5**) or may be remote from the surgical controller **62** (e.g., mounted on a stand with the tracking system **56**). The display device **64** is preferably adjustable so that the user can position/reposition the display device **64** as needed during a surgical procedure. For example, the display device **64** may be disposed on an adjustable arm (not shown) or to any other location well-suited for ease of viewing by the user. As shown in FIG. **5** there may be more than one display device **64** in the surgical system **50**.

(65) The display device **64** may be used to display any information useful for a medical procedure, such as, for example, images of anatomy generated from an image data set obtained using conventional imaging techniques, graphical models (e.g., CAD models of implants, instruments, anatomy, etc.), graphical representations of a tracked object (e.g., anatomy, tools, implants, etc.), constraint data (e.g., axes, articular surfaces, etc.), representations of implant components, digital or video images, registration information, calibration information, patient data, user data, measurement data, software menus, selection buttons, status information, and the like.

(66) In addition to the display device **64**, the computing system **52** may include an acoustic device (not shown) for providing audible feedback to the user. The acoustic device is connected to the surgical controller **62** and may be any known device for producing sound. For example, the acoustic device may comprise speakers and a sound card, a motherboard with integrated audio support, and/or an external sound controller. In operation, the acoustic device may be adapted to convey information to the user. For example, the surgical controller **62** may be programmed to signal the acoustic device to produce a sound, such as a voice synthesized verbal indication "DONE," to indicate that a step of a surgical procedure is complete. Similarly, the acoustic device may be used to alert the user to a sensitive condition, such as producing a tone to indicate that a surgical cutting tool is nearing a critical portion of soft tissue.

(67) To provide for other interaction with a user, embodiments of the subject matter described in this specification can be implemented on a computer having input device **66** that enables the user to communicate with the surgical system **50**. The input device **66** is connected to the surgical controller **62** and may include any device enabling a user to provide input to a computer. For example, the input device **66** can be a known input device, such as a keyboard, a mouse, a trackball, a touch screen, a touch pad, voice recognition hardware, dials, switches, buttons, a trackable probe, a foot pedal, a remote control device, a scanner, a camera, a microphone, and/or a joystick. For example, input device **66** can allow the user to make the selections as described above to adjust the surgical plan. Other kinds of devices can be used to provide for interaction with a user as well; for example, feedback provided to the user can be any form of sensory feedback, e.g., visual feedback, auditory feedback, or tactile feedback; and input from the user can be received in any form, including acoustic, speech, or tactile input. In addition, a computer can interact with a user by sending documents to and receiving documents from a device that is used by the user; for example, by sending web pages to a web browser on a user's client device in response to requests received from the web browser.

(68) The system **50** also includes a tracking (or localizing) system **56** that is configured to determine a pose (i.e., position and orientation) of one or more objects during a surgical procedure to detect movement of the object(s). For example, the tracking system **56** may include a detection device that obtains a pose of an object with respect to a coordinate frame of reference of the detection device. As the object moves in the coordinate frame of reference, the detection device tracks the pose of the object to detect (or enable the surgical system **50** to determine) movement of the object. As a result, the computing system **52** can capture data in response to movement of the tracked object or objects. Tracked objects may include, for example, tools/instruments, patient anatomy, implants/prosthetic devices, and components of the surgical system **50**. Using pose data from the tracking system **56**, the surgical system **50** is also able to register (or map or associate) coordinates in one space to those in another to achieve spatial alignment or correspondence (e.g., using a coordinate transformation process as is well known). Objects in physical space may be registered to any suitable coordinate system, such as a coordinate system being used by a process running on the surgical controller **62** and/or the computer device of the haptic device **54**. For example, utilizing pose data from the tracking system **56**, the surgical system **50** is able to associate the physical anatomy, such as the patient's spine, with a representation of the anatomy (such as an image displayed on the display device **64**). Based on tracked object and registration data, the surgical system **50** may determine, for example, a spatial relationship between the image of the anatomy and the relevant anatomy.
(69) Registration may include any known registration technique, such as, for example, image-to-image registration (e.g., monomodal registration where images of the same type or modality, such as fluoroscopic images or MR images, are registered and/or multimodal registration where images of different types or modalities, such as MRI and CT, are registered); image-to-physical space registration (e.g., image-to-patient registration where a digital data set of a patient's anatomy obtained by conventional imaging techniques is registered with the patient's actual anatomy); and/or combined image-to-image and image-to-physical-space registration (e.g., registration of preoperative CT and MRI images to an intraoperative scene). The computing system **52** may also include a coordinate transform process for mapping (or transforming) coordinates in one space to those in another to achieve spatial alignment or correspondence. For example, the surgical system **50** may use the coordinate transform process to map positions of tracked objects (e.g., patient anatomy, etc.) into a coordinate system used by a process running on the computer of the haptic device and/or the surgical controller **62**. As is well known, the coordinate transform process may include any suitable transformation technique, such as, for example, rigid-body transformation, non-rigid transformation, affine transformation, and the like.
(70) The tracking system **56** may be any tracking system that enables the surgical system **50** to continually determine (or track) a pose of the relevant anatomy of the patient. For example, the tracking system **56** may include a non-mechanical tracking system, a mechanical tracking system, or any combination of non-mechanical and mechanical tracking systems suitable for use in a surgical environment. The non-mechanical tracking system may include an optical (or visual), magnetic, radio, or acoustic tracking system. Such systems typically include a detection device adapted to locate in predefined coordinate space specially recognizable trackable elements (or trackers) that are detectable by the detection device and that are either configured to be attached to the object to be tracked or are an inherent part of the object to be tracked. For example, a trackable element may include an array of markers having a unique geometric arrangement and a known geometric relationship to the tracked object when the trackable element is attached to the tracked object. The known geometric relationship may be, for example, a predefined geometric relationship between the trackable element and an endpoint and axis of the tracked object. Thus, the detection device can recognize a particular tracked object, at least in part, from the geometry of the markers (if unique), an orientation of the axis, and a location of the endpoint within a frame of reference deduced from positions of the markers.

(71) The markers may include any known marker, such as, for example, extrinsic markers (or fiducials) and/or intrinsic features of the tracked object. Extrinsic markers are artificial objects that are attached to the patient (e.g., markers affixed to skin, markers implanted in bone, stereotactic frames, etc.) and are designed to be visible to and accurately detectable by the detection device. Intrinsic features are salient and accurately locatable portions of the tracked object that are sufficiently defined and identifiable to function as recognizable markers (e.g., landmarks, outlines of anatomical structure, shapes, colors, or any other sufficiently recognizable visual indicator). The markers may be located using any suitable detection method, such as, for example, optical, electromagnetic, radio, or acoustic methods as are well known. For example, an optical tracking system having a stationary stereo camera pair sensitive to infrared radiation may be used to track markers that emit infrared radiation either actively (such as a light emitting diode or LED) or passively (such as a spherical marker with a surface that reflects infrared radiation). Similarly, a magnetic tracking system may include a stationary field generator that emits a spatially varying magnetic field sensed by small coils integrated into the tracked object.
(72) The haptic device **54** may be the Tactile Guidance System™ (TGS™) manufactured by MAKO Surgical Corp., and used to prepare the surface of the patient's bone for insertion of the spinal plate **10**. The haptic device **54** provides haptic (or tactile) guidance to guide the surgeon during a surgical procedure. The haptic device is an interactive surgical robotic arm that holds a surgical tool (e.g., a surgical burr) and is manipulated by the surgeon to perform a procedure on the patient, such as cutting a surface of a bone in preparation for spinal plate installation. As the surgeon manipulates the robotic arm to move the tool and sculpt the bone, the haptic device **54** guides the surgeon by providing force feedback that constrains the tool from penetrating a virtual boundary.
(73) The construction and arrangement of the systems and methods as shown in the various exemplary embodiments are illustrative only. Although only a few embodiments have been described in detail in this disclosure, many modifications are possible. Accordingly, all such modifications are intended to be included within the scope of the present disclosure. The order or sequence of any process or method steps may be varied or re-sequenced according to alternative embodiments. Other substitutions, modifications, changes, and omissions may be made in the design, operating conditions and arrangement of the exemplary embodiments without departing from the scope of the present disclosure.

Claims

1. A method comprising: determining planned implant placements for a plurality of surgical procedures using a planning algorithm; recording virtual pre-operative user modifications of the planned implant placements; generating an altered planning algorithm based on the virtual pre-operative user modifications; and planning an implant placement for a subsequent surgical procedure using the altered planning algorithm; and operating a surgical robot based on the implant placement planned for the subsequent surgical procedure.
2. The method of claim 1, wherein the planning algorithm and the altered planning algorithm output different implant positions relative to a bone model given a same set of inputs.
3. The method of claim 1, wherein the planned implant placements are positions of virtual implant models relative to virtual bone models.
4. The method of claim 1, wherein the virtual pre-operative user modifications comprise rotations of virtual implant models relative to virtual bone models.
5. The method of claim 1, wherein operating the surgical robot based on the implant placement for the subsequent surgical procedure.
6. The method of claim 1, further comprising providing transparency into operation of the altered planning algorithm by guiding, via display of a graphical user interface on a screen and based on

the implant placement for the subsequent surgical procedure, a user to provide a sequence of inputs to the graphical user interface that result in the graphical user interface displaying movement of a virtual implant from an initial placement to the implant placement for the subsequent surgical procedure.

7. The method of claim 1, comprising using a characteristic of a patient's joint as an input to the altered planning algorithm.

8. The method of claim 1, comprising using a presence of a type of bone structure as an input to the altered planning algorithm.

9. The method of claim 1, comprising using a physical characteristic of a patient as an input to the altered planning algorithm.

10. The method of claim 1, comprising storing the virtual pre-operative user modifications from a plurality of planning systems.

11. A method comprising: recording pre-operative user inputs that adjust planned placements of virtual implants relative to virtual bone models corresponding to a plurality of surgical procedures; updating an automated planning algorithm based on the pre-operative user inputs; automatically providing a planned implant position for a particular patient by executing the updated, automated planning algorithm on input information relating to the particular patient; and causing an actual implant to be installed in the particular patient in the planned implant position.

12. The method of claim 11, wherein the pre-operative user inputs are provided to a plurality of planning systems.

13. The method of claim 11, wherein a subset of the pre-operative user inputs cause rotations and translations of the virtual implants.

14. The method of claim 11, further comprising providing transparency into operation of the automated planning algorithm by guiding, by displaying a graphical user interface on a screen and based on the planned implant position for the particular patient, a user to provide inputs to the graphical user interface that result in the graphical user interface displaying the planned implant position.

15. The method of claim 11, wherein causing the actual implant to be installed in the particular patient comprises controlling a surgical robot to guide a resection that prepares the particular patient to receive the actual implant in the planned implant position.

16. The method of claim 11, wherein the input information indicates a physical characteristic or bone structure of the particular patient.

17. One or more non-transitory memory devices storing instructions that, when executed by one or more processors, causes the one or more processors to perform operations comprising: storing pre-operative user inputs that adjust planned placements of virtual implants relative to virtual bone models corresponding to a plurality of surgical procedures; updating an automated planning algorithm as a function of the pre-operative user inputs; and automatically providing a planned implant position for a particular patient by executing the updated, automated planning algorithm on input information relating to the particular patient, wherein providing the planned implant position comprises providing transparency into operation of the automated planning algorithm by guiding, via a graphical user interface on a screen and based on the planned implant position for the particular patient, a user to provide inputs to the graphical user interface that result in moving of a virtual implant to the planned implant position.

18. The one or more non-transitory memory devices of claim 17, wherein the pre-operative user inputs are received via a plurality of planning systems.

19. The one or more non-transitory memory devices of claim 17, wherein the input information relates to a bone structure of the particular patient.